Note: Assignment 5 is due on Friday 29 May 2026, 3:00 pm. No late submission without prior approval from the lecturer is accepted.

Assignment 5 – Q.1: (30 marks)

Develop a suitable strut-and-tie model for each of the following cases. Give detailed justification for your developed strut-and-tie model.

(i) A simply supported beam carrying a uniformly distributed load (Figure 1). (10 marks)

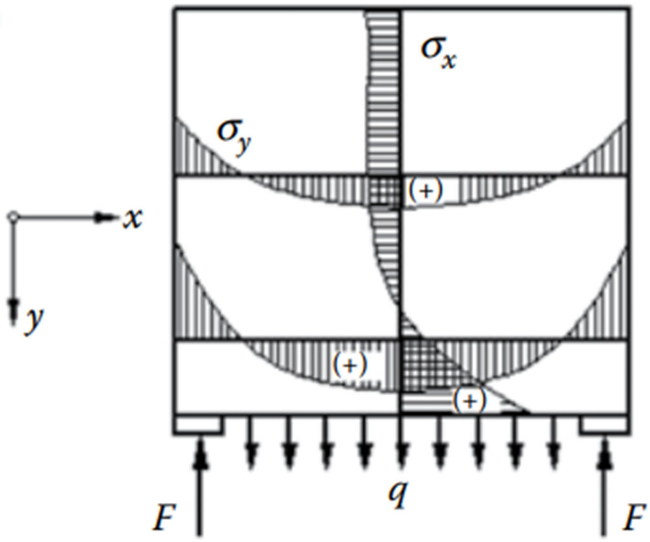

Figure 1. A simply supported beam carrying a uniformly distributed load.

(ii) A simply supported deep beam with opening (Figure 2). The beam width is 400 mm. (20 marks)

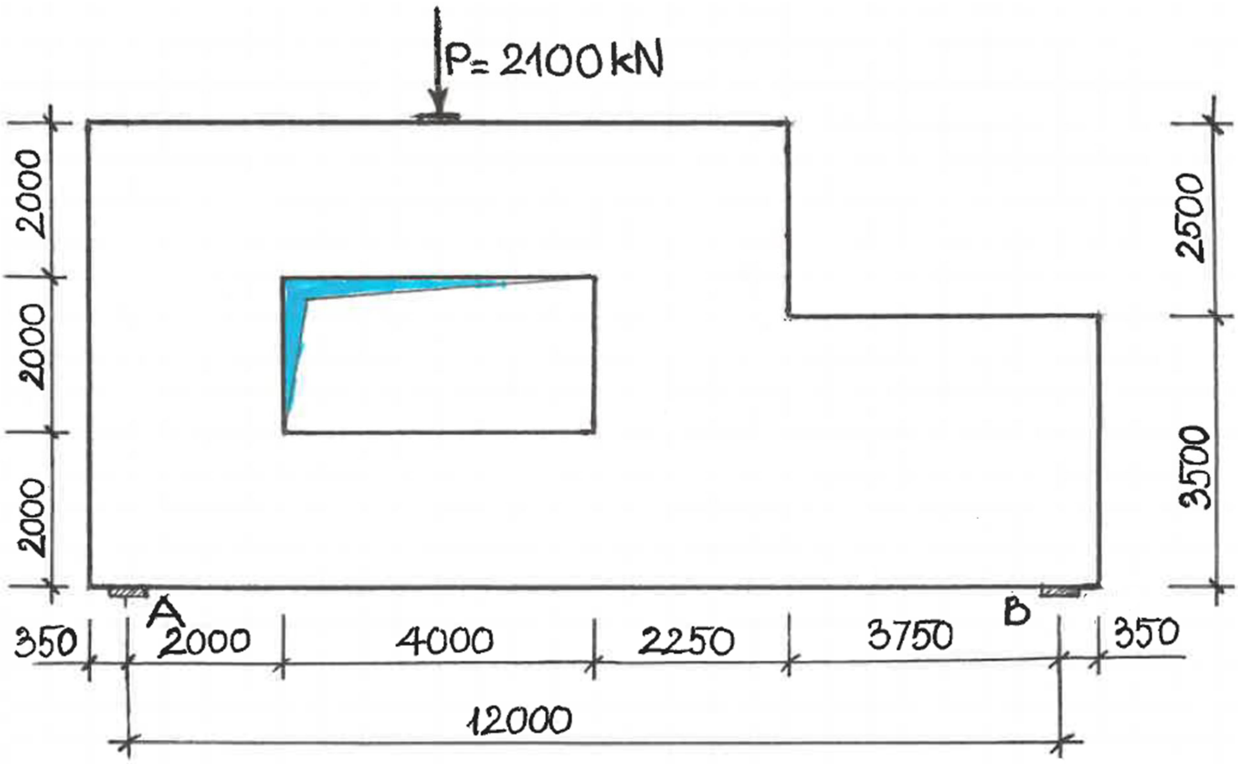

Figure 2. A simply supported deep beam with opening and varying depth.

Assignment 5 – Q.2: (70 marks)

On the basis of an appropriate strut-and-tie model, design the deep beam (Figure 3). Relevant details as follows:

- The width of all members (including beam and columns) is 600 mm.
- The factored design load P = 4 × m + n (kN); where m and n are the first 3 digits and the last 2 digits of your 8-digit student number, respectively. For example, if your student number is 49012345, your P value will be: P= 4 × 490 + 45 = 2005 kN.
- f'_c= 50 MPa; $f_{sy} = f_{sy.f}$ = 500 MPa.
- Clear cover to reinfrocement= 30 mm.
- Using relevant clauses in AS3600-2018.
- If your student number is an even number, your allocated support in questions ***c-d*** is the left support. If your student number is an odd number, your allocated support in questions ***c-d*** is the right support.

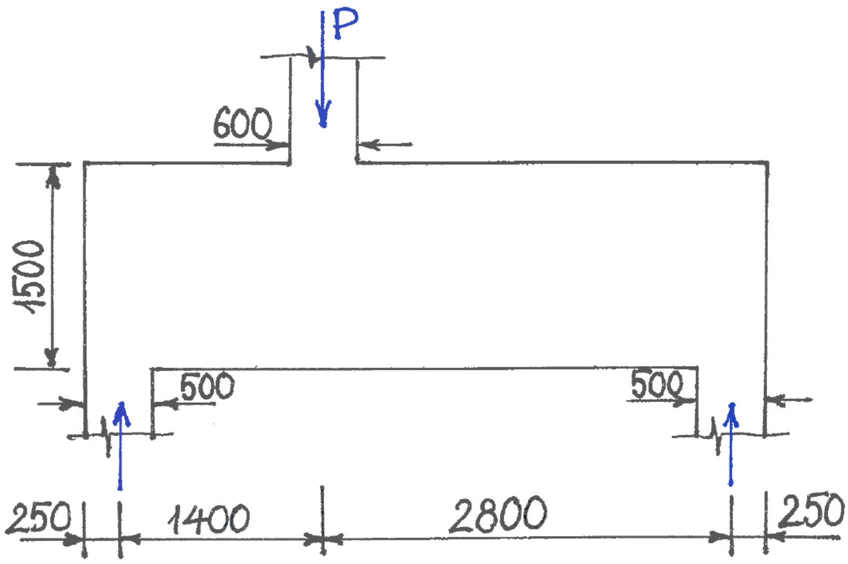

Figure 3. A simply supported beam carrying a point load.

You are required to:

a. Identify the B- and D- regions. Develop a suitable strut-and-tie model. Give justification for your chosen strut-and-tie model. (15 marks)

b. Select steel reinforcement for each of the tension ties. (10 marks)

c. Check stress levels in all struts and in the node at your allocated support. Using Mohr's circles to determine the maximum compressive stress in your allocated nodal zone when checking the stress level in that node. (30 marks)

d. Design for bursting reinfocement for the inclined strut coming into your allocated support. Assuming the strut force at serviceability limit state is 0.6 of that at ultimate limit state. (15 marks)